Nafs-Aql v0.1

Mathematical validation and technical report of a dual-process agent on Qwen

System 1 (fast) · System 2 (slow) · Active-Intellect verifiers · value core · gated distillation

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Architecture review of the Technical Specification, Model Choices note and Validation Addendum

18 October 2026

Outline

Part I – Problem and architecture

- Executive verdict
- Concept map and data flow
- Qwen roster and serving
- Formal setting and assumptions

Part II – Validation (12 results)

- Verifier, best-of-N, cascade cost
- Router, consensus, value core
- Coupling, self-model, measurement floors
- Distillation, serving, end-to-end identity

Part III – Technical report

- Defect register and amended design
- Features and capabilities
- Revised constants and gates
- Development plan, risks, open items

Method

Every numeric claim of the addendum was recomputed independently (all reproduced). Extra results added here: heterogeneity gap for T2 and the cascade difference identity (P1).

Part I: Problem and architecture

Executive verdict

Verdict

Buildable on Qwen: yes, with amendments.
The structure is sound: cheap policy → calibrated router → verified best-of-N → scoring judge → metrics-only self-model → gated distillation.

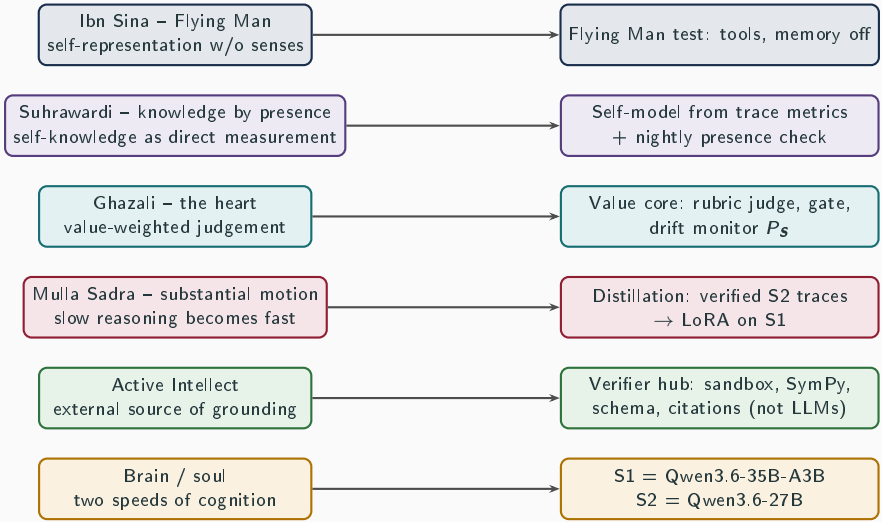
What holds

- Roster is implementable (embedding/reranker versions and MoE-LoRA are the open risks)
- Verifier-grounded loop is the right core
- Metrics-only self-model blocks confabulation
- Human-signed promotion limits drift

Ten defects found in text, code and thresholds

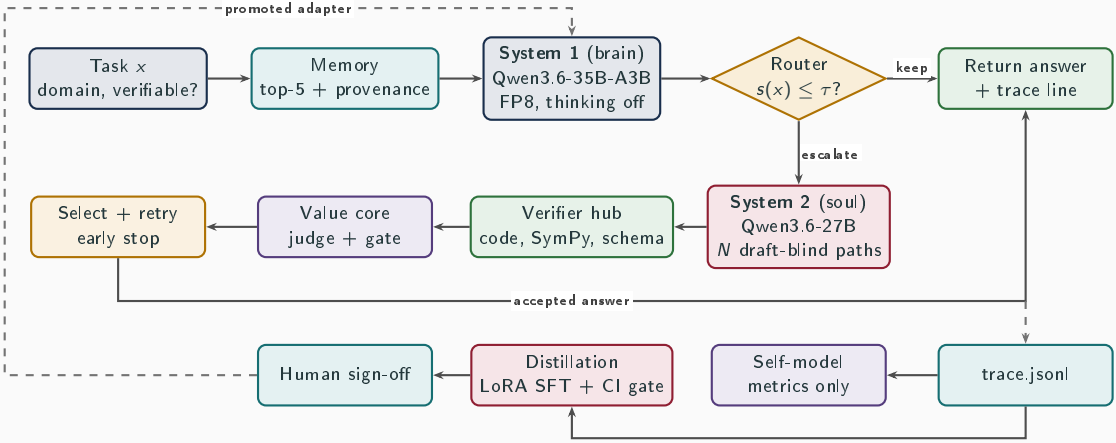
1. Cascade beats S2-only only if escalation $\lesssim 5\%$ (T3); baseline undefined
2. 500-task baseline too small for most gates
3. Hard gate not coded; vacuous hedge scores 0.75
4. $V_{\min}$=0.9 lets code fail 10% of tests; consensus $\Rightarrow$ unanimity
5. τ adapted to target g, yet g is the learning diagnostic
6. Self-model biased by routing selection
7. ECE gate 0.08 unmeasurable at n=500
8. S1 draft passed to S2 invites anchoring
9. Math verifier lists LLM rederivation
10. Router pseudo-code $\neq$ prose rule

Concept map: from philosophy to modules



Metaphor only: the project makes no claim of consciousness. The formal content is in Part II.

System architecture and data flow



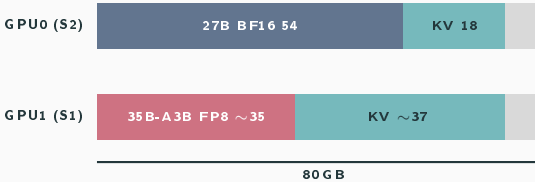
Qwen roster and serving profile

Role	Model	Mode / status
System 1	Qwen3.6-35B-A3B (MoE, ∼3B active)	thinking off; official FP8 repo; card details to confirm
System 2	Qwen3.6-27B dense, Apache 2.0, 262K ctx	thinking on, `preserve_thinking`
Value core	same 27B weights	thinking off, rubric, $T=0$
Embed / rerank	Qwen3-Embedding-4B, Reranker-4B	**unverified**; read dim from config
Verifiers	sandbox, SymPy, JSON-Schema	deterministic, non-LLM

Pin and re-baseline

Pin exact revisions. Any change of model, template or adapter re-runs the Phase 0 baseline. Do not chase Qwen3.7/3.8/4 mid-build.

Profile A memory budget (80 GB GPUs, util 0.90)



Profile B (48 GB, 4-bit) is development-only: no benchmarking.

Per-request switches

S1: `enable_thinking=False`
S2: `enable_thinking=True, preserve_thinking=True`
S1 served without a reasoning parser.

Formal setting

Definition (Objects)

Task x; C_1, C_2 = events that the S1 draft / S2 answer is correct. A verifier accepts a correct answer w.p. r (true-accept) and a wrong one w.p. φ (false-accept). $q(x) = \mathrm{P}(\text{one S2 path correct})$; N paths, R retries, k agreement samples; c_1, c_p = cost of one S1 call / S2 path; g = escalation rate; $s(x)$ = router score.

Goal

Maximise verified accuracy subject to expected cost, i.e. min $\mathbb{P}[\text{final wrong}] + \lambda\,\mathbb{E}[\text{cost}]$ over router π, selector σ, and S1 weights θ (via distillation).

Assumptions

- **A1** S2 paths conditionally i.i.d. given x
- **A2** (r, φ) identical for every candidate source
- **A3** verifiers see only task + final answer
- **A4** held-out data $\sim$ deployment

Weak points: A1 and A2 (shared weights, Goodhart pressure). Each result states what breaks if they fail.

Part II: Validation

T1 – Precision of a verified return

Theorem (T1)

Prior correctness p. Then

$$\rho = \mathbb{P}[\textit{wrong} \mid \textit{accepted}] = \frac{\varphi(1-p)}{\varphi(1-p) + rp}.$$

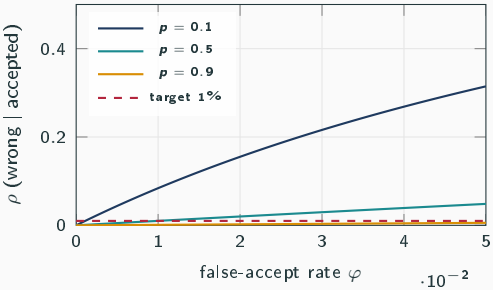
$$\rho \le \varepsilon \iff \varphi \le \frac{\varepsilon\, r\, p}{(1-\varepsilon)(1-p)}.$$

Proof. Bayes' rule. □

Consequence. Hard tasks p=0.10, r=0.98, ε=0.01 need $\varphi \le 0.0011$; with φ=0.02, $\rho = 0.155$.

Design verdict

Grounding is limited by verifier φ, not by model strength. Every verifier must publish (r, φ) from *mutation testing*.



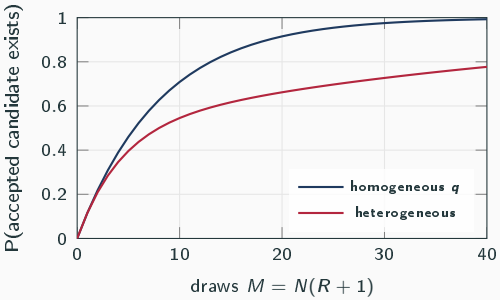
T2 – Best-of-N buys coverage, not precision

Theorem (T2)

$a = qr + (1-q)\varphi$. Uniform pick among accepted paths:
(i) $\mathbb{P}[\text{wrong} \mid \geq 1 \text{ accepted}] = (1-q)\varphi/a$ for every N;
(ii) coverage with $M = N(R+1)$ draws is $1-(1-a)^M$;
(iii) if q varies, coverage $\leq 1-(1-\mathbb{E}[a])^M$.

Proof. (i) accepted paths are i.i.d. wrong w.p. $(1-q)\varphi/a$; (ii) geometric; (iii) $u \mapsto (1-u)^M$ convex, Jensen. □

Example (q=.1): error stays 15.5%; coverage 62.7% at M=8, 94.8% at M=24 if homogeneous.
Added check: half the tasks q=0, half q=0.2 (same $\mathbb{E}[a]$) gives only 69% at M=24.



Amendment: choose N from the coverage target; select by score only if it carries information beyond the pass bit; stop at first accepted path on verifiable tasks.

T3 – Cascade cost and break-even escalation

Theorem (T3)

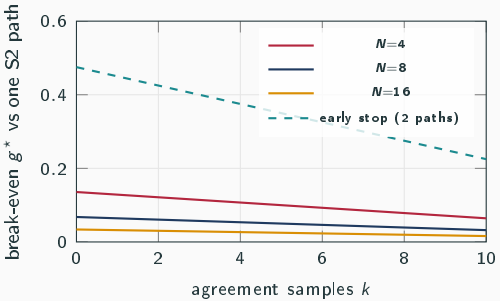
Let $s = P(\text{a round contains an acceptable candidate})$, $E = \frac{1-(1-s)^{R+1}}{s}$. Expected cost $C = (1+k)c_1 + v + g\,Nc_pE$. The cascade beats baseline cost B iff

$$g < \frac{B - (1+k)c_1 - v}{Nc_pE}.$$

Proof. Linearity; truncated geometric sum. □

Numbers (c_1=.05c_p, k=4, N=8, s=.5, E=1.75):

- vs one S2 path: $g < 5.4$–5.7%
- vs same N-path loop (B=14c_p): $g < 98\%$
- early stop on verifiable tasks: about 2 draws, not 8



Amendment: define the Phase-1 baseline (S2 with the same loop, or report the cost–accuracy frontier); use n=4 in one request for the k samples.

T4 – The router is a threshold rule

Theorem (T4)

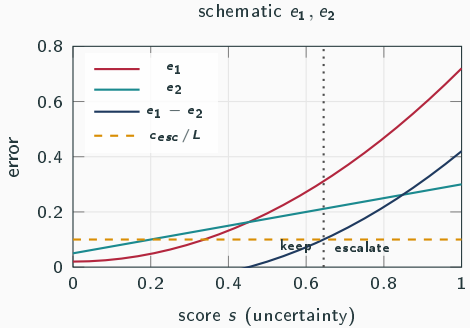
Let $e_1(s) = \mathbb{P}[S1\ wrong \mid s]$, $e_2(s) = \mathbb{P}[final\ wrong\ after\ S2 \mid s]$, L = *error cost*, c_{esc} = *extra escalation cost. Optimal: escalate iff* $e_1(s) - e_2(s) > c_{esc}/L$; *a single threshold* τ *if the gap is monotone.*

Proof. Answer: Le_1. Escalate: $c_{esc} + Le_2$. Compare. □

Quantile adaptation (DKW). Realised g within ε of $g^\star$ w.p. $1-\delta$ if $n \geq \ln(2/\delta)/(2\varepsilon^2)$: n=512 $\Rightarrow \varepsilon$=.06; ε=.02 needs 4,611.

Diagnostic conflict

If τ tracks a target g, then g is constant by construction and cannot show learning. Fix τ (cost formula) and watch g fall, *or* fix g and watch S1 pass rate m rise.



Amendments: one calibrated score (logistic on labelled tasks) instead of separate H, A; estimate e_2 (correlated with e_1, same family); `mean_entropy` is a truncated top-5 proxy, so 0.35 nats is not meaningful until calibrated.

T5 – Consensus arithmetic and its limit

Theorem (T5)

consensus$(c) = k/(N-1)$, k = *other candidates agreeing.* *consensus* $\geq \theta \iff k \geq \lceil \theta(N-1) \rceil$.
N=8: θ=.75 *needs 7 of 8 agreeing;* θ=.9 *needs 8 of 8.*
N=4: *both need 4 of 4.*
$\mathbb{P}$[*unanimous and wrong*] $\leq p_{\max}^{N-1}(1-q)$ *per task.*

Limit

Averaged over tasks this tends to the mass of tasks where the model *deterministically* errs. It does not vanish as N grows.

Consequences

- $V_{\min}$=0.9 applied to consensus silently means unanimity
- Express consensus in candidate counts
- Keep `assurance=consensus` flagged
- Never write memory or training data from consensus without human check
- Pseudo-code `ceil(theta*(N-1))+1` must state whether `agree_count` includes the candidate itself

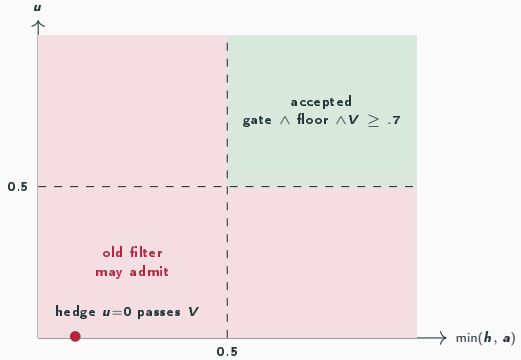
T6 – Value score algebra

Theorem (T6)

$V = 0.5 \min(h, a) + 0.25u + 0.25m$ *on* $[0, 1]^4$.
(i) $V \geq .7$ *does not imply the gate:* h=.49, a=u=m=1 *gives* .745.
(ii) Vacuous hedge h=a=m=1, u=0 *gives* $.75 \geq V_{\min}$.

Proof. Substitute. □

Amendment. Explicit gate $h, a \geq .5$ and helpfulness floor $u \geq .5$, plus $V \geq .7$. Per-verifier $V_{\min}$: 1.0 for exact verifiers (code, schema), 0.9 for soft ones.



T7 – Coupling between System 1 and System 2

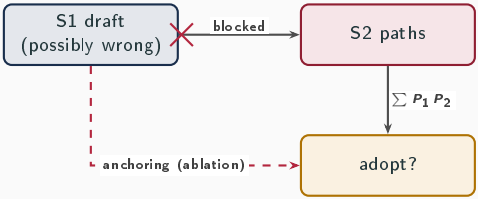
Theorem (T7)

If `sample_paths` *does not read the S1 draft, then*

$$\mathbb{P}[\textit{S2 adopts wrong draft} \mid \textit{draft wrong}] = \sum_{a \in \textit{wrong}} P_1(a) P_2(a),$$

the collision probability of two independent answer laws. Passing the draft adds anchoring on top. Certifying adoption $< 5\%$ at 95% with zero observed needs $n \geq \ln .05 / \ln .95$, i.e. 59 trials; with f adoptions use Clopper–Pearson.

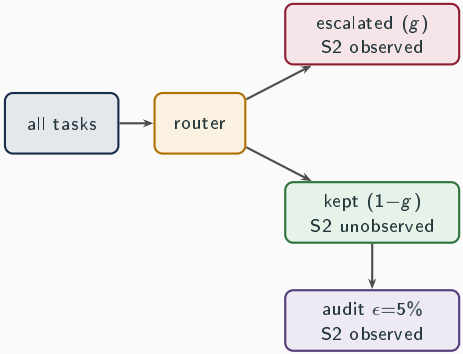
Amendment: System 2 is *draft-blind* by default; draft-conditioned mode is an ablation; run the coupling test on both with ≥ 200 trials. Same model family $\Rightarrow$ collision probability is not small: measure it.



T8 – Self-model bias under routing

Theorem (T8)

s2_pass is observed only on escalated tasks, i.e. those where S1 failed: the naive estimate is $\mathbb{P}[\textit{correct} \mid \textit{S1 failed}]$. *Audit a random fraction* ϵ *of non-escalated tasks with S2. Then* $\hat{p}_2 = g\,\hat{p}_{2|esc} + (1-g)\,\hat{p}_{2|audit}$ *is unbiased (IPW, positivity* $\epsilon > 0$*), with* $\mathrm{SE} \le \sqrt{\frac{g^2 p(1-p)}{ng} + \frac{(1-g)^2 p(1-p)}{n(1-g)\epsilon}}$.



$n{=}10^4, g{=}.3, \epsilon{=}.05$: 350 audited, $\mathrm{SE} \le 0.019$.

Also: non-verifiable domains are *unmeasured*, not 0.0; key stats by adapter hash with a sliding window.

T9, T10 – Measurement floors of the evaluation suite

Gate	Spec	Problem	Needed
Escalation rate	500	$\pm$6.1 pts	4,611 for $\pm$2
ECE $\leq$ .08	500, 10 bins	noise floor 0.034	5,700 (10 bins) / 2,900 (5) or debias
Presence $\leq$ 10%	100	obs. 10% $\Rightarrow$ UB .16	$\geq$ 500
Flying Man $\geq$ .90	30	27/30 $\Rightarrow$ LB .76	$\geq$ 60
Coupling $\leq$ 5%	not stated	59 with 0 hits	200
P1 non-inferior (1 pt)	500	resolves 3.5 pts	6,200 pairs (δ=.1); 690 for 3 pts
Drift	K=50, α=.05	2.6 false alarms/yr	$K \geq (2.487\sigma/\Delta)^2$; α=.01/CUSUM

Formulas

ECE bias for calibrated outputs: $\mathbb{E}[\widehat{\mathrm{ECE}}] \approx \sqrt{\frac{2}{\pi}}\sqrt{\bar{p}(1-\bar{p})}\sqrt{B/n}$.

Non-inferiority: $n \geq (z_\alpha + z_\beta)^2 \delta / m^2$, with $z_\alpha + z_\beta = 2.487$.

Drift: effective sample is K (probe pairs correlated); σ=.1 $\Rightarrow$ $K \geq 25$; σ=.2 $\Rightarrow$ $K \geq 99$.

Add a *negative control* (shuffled/contradictory commitments): embedding cosine measures wording more than stance.

T11 – Distillation: noise, measured gain, gate size

Theorem (T11)

(a) If a fraction η of labels is wrong, $|\mathrm{err}_{noisy}(h) - \mathrm{err}_{clean}(h)| \le \eta$ *for every h; here* $\eta = \rho$ *of T1.*
(b) Eval verifier (r, φ)*: measured rate* $= \varphi + (r - \varphi)t$*, so a measured +2 pts is a true* $+2/(r - \varphi)$ *pts; training on verifier-accepted outputs can raise* φ *(Goodhart) and inflate it.*
(c) Paired gate, discordance δ*:* $n \ge z^2\delta/\varepsilon^2$*.* δ=.1, ε=.01*: 2,706 per domain; Bonferroni over 5 (z*=2.326*): 5,410.*

Proof (a). The events differ only where the noisy label is wrong. □

Amended promotion gate

- lower bound of overall gain > 0, point estimate ≥ 2 pts
- lower bound of each domain change ≥ -1 pt
- eval on a verifier never used in collection
- keep 5% abstain examples
- ECE, Flying Man, presence still pass
- human sign-off; hash into self-model

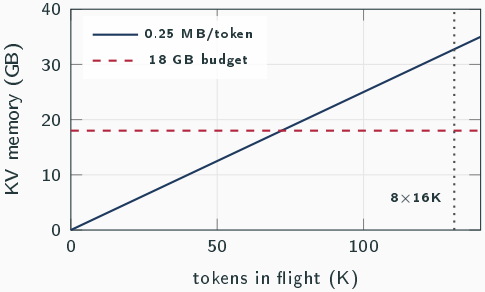
T12 – Serving capacity

Theorem (T12)

KV bytes per sequence $= 2\, L_{kv}\, H_{kv}\, d_{head}\, b\, T$, *where* L_{kv} *counts layers that keep a KV cache. S2 tokens in flight* $= N(\mathit{prompt} + \mathit{generated})$.

Illustration (hypothetical dims; read `config.json`): 64 layers, 8 KV heads, d=128, 2 B $\Rightarrow$ 0.25 MB/token. GPU0 leaves $0.9 \cdot 80 - 54 = 18$ GB $\approx$ 72K tokens; 8 paths $\times$ 16K = 131K tokens, so paths queue or are preempted.

Amendment. Derive concurrency from the real config (a hybrid-attention model shrinks L_{kv}), then budget N and `max_tokens`, or lower N.



P1 – End-to-end identity (added): when is the cascade safe?

Proposition (Cascade difference identity)

Let $\rho_1 = P(\text{wrong} \mid \text{router keeps the S1 answer})$, and $\bar{e}_2^{\,kept} = $ error S2-only would make on those same tasks. If S2 behaves identically on escalated tasks in both systems, then

$$\mathbb{P}[\text{cascade wrong}] - \mathbb{P}[\text{S2-only wrong}] = (1 - g)\left(\rho_1 - \bar{e}_2^{\,kept}\right).$$

Proof. The two systems coincide on the escalated fraction g and differ only on the kept fraction $1 - g$. □

Corollaries

- Non-inferiority within margin m requires $\rho_1 \le \bar{e}_2^{\,kept} + m/(1-g)$; e.g. m=.01, g=.3: $\rho_1 \le \bar{e}_2^{\,kept} + 1.4$ pts.
- $\bar{e}_2^{\,kept}$ is observable only through audits (T8): the audit stream is part of the safety case, not telemetry.
- With a verifier on kept answers, ρ_1 is T1's ρ: control φ.

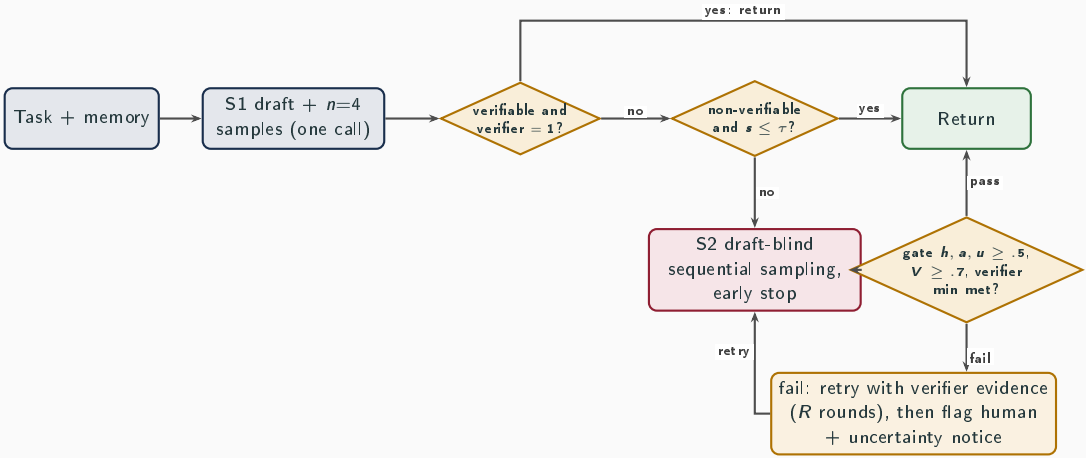


Part III: Technical report

Defect register and amendments

#	Defect	Result	Amendment
1	Cascade not cheaper vs same-loop S2 unless g small	T3	define baseline; early stop; n=4 agreement call
2	500-task baseline too small	T4,T9,T11	$\geq$4,600; 6,200 for 1-pt non-inferiority
3	Hard gate not coded; hedge passes	T6	gate + helpfulness floor $u \geq .5$
4	$V_{\min}$=.9 lets code fail 10%; consensus = unanimity	T5	1.0 for exact verifiers; consensus in candidates
5	τ adaptation vs g diagnostic	T4	fix cost threshold or fix g, not both
6	Capability table biased by routing	T8	audit ϵ=.05; mark unmeasured
7	ECE 0.08 unmeasurable at 500	T9	5 bins, $n \geq$ 2,900, or debiased ECE
8	S1 draft anchors S2	T7	draft-blind default; coupling $\geq$ 200
9	LLM rederivation in math verifier	–	deterministic checks only
10	Router code $\neq$ prose	T4	one calibrated score; state rule for failed verifiable

Amended decision flow



A verifiable task that fails verification is not eligible for the entropy shortcut and escalates (resolves defect 10). Every return is logged; 5% of kept tasks also get an S2 audit (T8).

Amended selection code (replaces the `ok` filter)

```
def accept(c, task):
    if c.honesty < 0.5 or c.harm_avoidance < 0.5 or c.helpfulness < 0.5:
        return False                          # hard gate + helpfulness floor
    if c.value_score < VAL_MIN:               # 0.7
        return False
    if task.verifiable:
        return c.verifier_score >= c.verifier_min   # 1.0 for exact verifiers
    # non-verifiable: count in candidates, flag assurance=consensus
    return c.agree_count >= math.ceil(THETA * (N - 1))
```

Selection among accepted: sort by calibrated score only if it adds information beyond the pass bit (T2); otherwise uniform.

Math verifier: SymPy or numeric equality against a reference. No LLM rederivation, preserving "verifiers are not LLMs".

Features and capabilities

Features

- Dual-process routing with calibrated score
- Verified best-of-N with early stop and evidence-fed retries
- Verifier hub: code sandbox, SymPy, JSON Schema, citation check; each with (r, φ)
- Rubric value core with hard gate and drift monitor
- Memory: episodic, semantic (verified only), procedural; provenance in prompt
- Metrics-only self-model; nightly presence check
- Gated weekly LoRA distillation of verified traces

Capabilities (what it can deliver)

- Verified answers on code, math, schema tasks, with precision bounded by T1
- Calibrated abstention and flagged uncertainty elsewhere
- Stable, measurable self-description
- Gradually cheaper operation as S1 absorbs verified skill

Honest limits

Non-verifiable domains stay at consensus assurance (T5); S1/S2/judge errors are correlated (shared family); no consciousness claim.

Revised constants

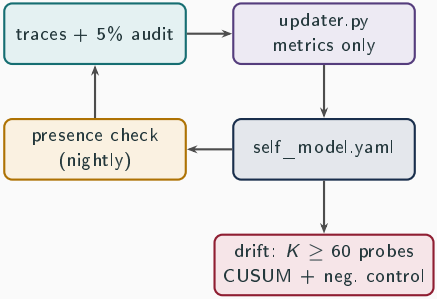
Parameter	Current	Proposed
Phase 0 baseline	500 tasks	$\geq$4,600; 6,200 for 1-pt non-inferiority (700 for 3 pts)
ECE	10 bins, n=500	5 bins with $n \geq 2{,}900$, or debiased
Presence questions	100	≥ 500
Flying Man probes	30	≥ 60
Coupling trials	not stated	≥ 200, draft-blind and conditioned
Code / schema V_{min}	0.9	1.0 (soft verifiers 0.9)
Helpfulness floor	none	0.5
Audit rate ϵ	none	0.05
Drift test	$\alpha = .05$	$\alpha = .01$ or CUSUM + negative control
Promotion gate	point estimates	confidence bounds, held-out verifier

All values remain functions of Phase 0 measurements, not fixed targets.

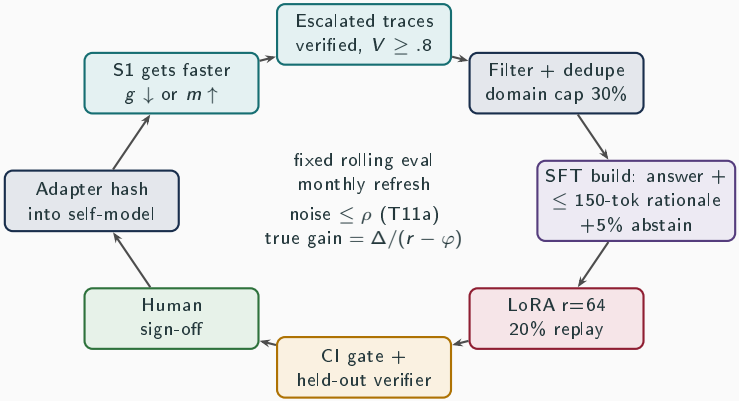
Self-model and value drift loop

```
capabilities:    # updater.py only
  code: {s1_pass: .., s2_pass: ..,
         ece: .., n: .., adapter: <hash>}
  math: {...}
  factual: unmeasured    # not 0.0
limits:          # from measured failures
goals:           # human-edited only
commitments:     # human-edited only
```

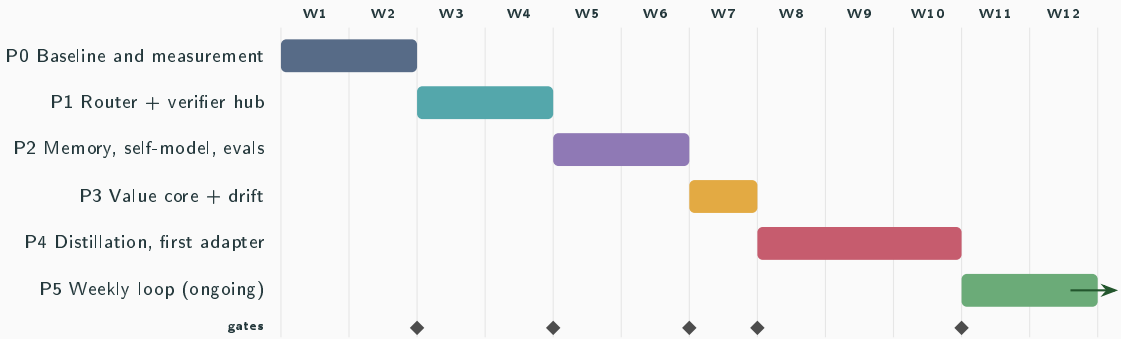
Sliding window keyed by adapter hash; S2 gets $\leq$2,000 tokens, S1 a 200-token digest.



Distillation loop (substantial motion)



Development plan (re-baselined)



Phase 0 grows from 1 to 2 weeks to reach the $\geq$4,600-task floor; later phases shift by one week (proposed). Each diamond is a gate on the next slide.

Phase gates (amended)

Phase	Deliverable	Acceptance
P0	Serve both models; $\geq$4,600-task multi-domain run of S1-only and S2-only	Stable endpoints; estimates of q, (r, φ), $e_1(s)$, $e_2(s)$, c_1/c_p, δ, KV capacity
P1	Router, verifier hub (code, math, schema) with mutation tests	Cost below S2 *with same loop*, accuracy loss within 1 pt at 95% (6,200 pairs) or 3 pts (700)
P2	Memory, self-model updater, audit stream, Flying Man, presence	Both tests end to end; $\geq$ 60 probes, $\geq$ 500 questions; coupling $\geq$ 200 trials
P3	Value core, drift monitor	Baseline P_S, measured σ, negative control separates, alerts fire
P4	Distillation pipeline, first adapter	Confidence-bound gate passes; human sign-off
P5	Weekly collect, train, gate, promote	Gate each cycle; weekly 2% human audit of judge

Risk register

Risk	Mechanism	Mitigation
Version drift	Qwen releases fast; embed/rerank unverified	Pin revisions and templates; re-baseline on any change
Correlated errors	S1, S2, judge share family (A1, A2 fail)	Independent verifiers; human audit; measure collision (T7)
Verifier false accepts	ρ set by φ (T1)	Mutation testing; per-verifier $V_{\min}$; private test set
Goodhart in distillation	φ rises when training on accepted outputs	Held-out verifier; confidence-bound gate (T11)
Self-model bias	Selection by routing (T8)	5% audit; unmeasured flag; adapter-keyed window
KV exhaustion	$N \times 16$K tokens exceed budget (T12)	Config-derived concurrency; lower N or tokens
Non-verifiable assurance	consensus has a floor (T5)	Flag; prefer abstain; human check before reuse
MoE LoRA tooling	expert-layer support uncertain	Attention-only LoRA fallback; dense S1 fallback

Conclusion and open items

Conclusion

Nafs-Aql is a coherent, buildable design on Qwen. Its soundness rests on three quantities that must be measured, not assumed: verifier false-accept φ, router calibration $e_1(s), e_2(s)$, and S1/S2 error correlation. With the ten amendments, every gate is both implementable and statistically resolvable.

Open items (Phase 0 outputs)

- q per domain; (r, φ) per verifier
- e_1, e_2; c_1/c_p; discordance δ; drift σ
- real KV capacity from `config.json`
- confirm 35B-A3B card, embedding/reranker versions, any newer Qwen release